

HG002 BENCHMARK-AWARE SUBSTRATE VALIDATION

VAP Upstream Variant Substrate Benchmarking Against GIAB HG002 v4.2.1 (GRCh38) Using hap.py

BENCHMARK-AWARE • REPRESENTATION-AWARE • PROVENANCE-AWARE • REPRODUCIBLE • ENGINEERING TRUST



BENCHMARK ARCHITECTURE

- GRCh38 Benchmarking (Primary Assembly)
- GIAB HG002 v4.2.1 (BED + VCF)
- High-Confidence BED Restriction
- hap.py Evaluation
- Containerized Execution (Apptainer)
- Representation-Aware Comparison

NAMESPACE MEDIATION

Rule: Strip leading 'chr' prefix from GIAB truth contigs

Original (GIAB)	Harmonized (Run-Scoped)
chr1	1
chr2	2
:	:
chr22	22

Canonical GIAB files remain unchanged

REPRESENTATION-AWARE BENCHMARKING



Representation-Aware Evaluation

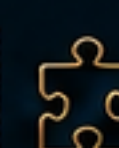


Validates Upstream Substrate Boundary

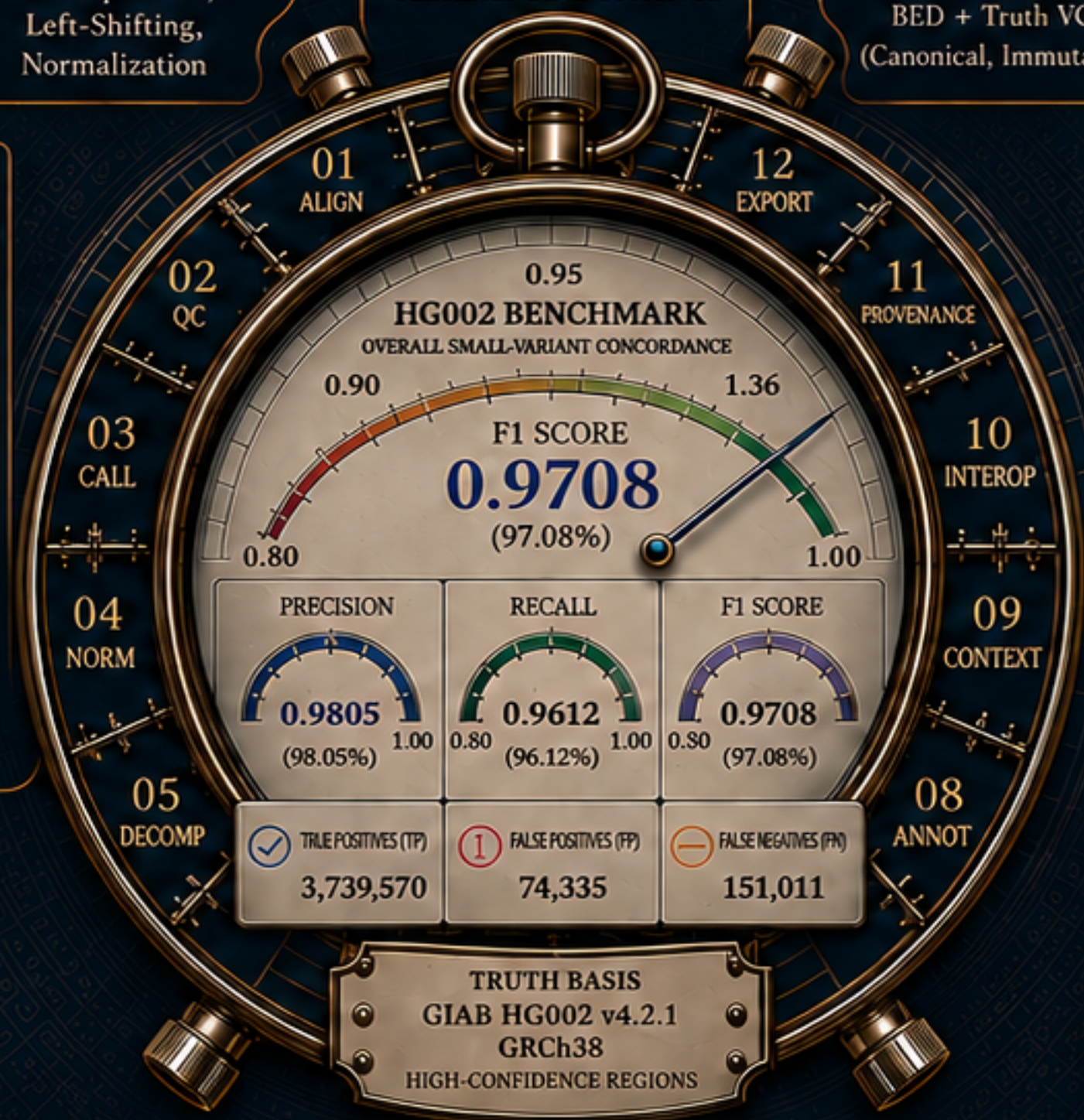
Benchmarking validates substrate integrity, not downstream biological interpretation.



Engineering Trust



Biological Trust



MEASURED CONCORDANCE (ALL VARIANTS)

True Positives (TP)	3,739,570
False Positives (FP)	74,335
False Negatives (FN)	151,011

VARIANT-CLASS PERFORMANCE

SNPs			INDELs		
Precision	0.9856	(98.56%)	Precision	0.9457	(94.57%)
Recall	0.9790	(97.90%)	Recall	0.8472	(84.72%)

All metrics computed within GIAB high-confidence regions.

REPRODUCIBILITY & PROVENANCE

Run ID	run_2026_06_03,010030
Runtime Environment	Apptainer (Containerized)
Benchmark Tool	hap.py (v0.3+)
Generated	2026-06-03T04:22:12Z
Provenance	Immutable Artifacts
Reproducible	Deterministic Execution

KEY TAKEAWAY

- Strong small-variant concordance against GIAB HG002 v4.2.1
- Benchmarking restricted to GIAB high-confidence regions
- Representation-aware normalization preserves equivalence handling
- Provenance-aware execution establishes **engineering trust**
- Benchmarking validates substrate integrity, not biological interpretation

DISCIPLINED BENCHMARKING. MEASURED CONCORDANCE. ENGINEERING TRUST.